

# BML lecture #3: Alternatives to MCMC

<http://github.com/rbardenet/bml-course>

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**2** Variational Bayes

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**4** Beyond VB

## 1 Introduction

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## 4 Beyond VB

What comes to *your* mind when you hear "variational inference"?

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## Turning integration into optimization over measures

Variational Bayesian inference (VB) consists in approximating

$$\int f(\theta)\pi(\theta)d\theta \approx \int f(\theta)q^*(\theta)d\theta$$

with  $q^* \in \arg \min_{q \in \mathcal{Q}} \text{distance}(\pi, q)$ . Often we take

$$\text{distance}(\pi, q) = \text{KL}(q, \pi) := \int q(\theta) \log \frac{q(\theta)}{\pi(\theta)} d\theta.$$

for computational convenience.

But remember we can only evaluate  $\pi_u = Z\pi\ldots$

► Show that  $J(q) := \int q(\theta) \log \frac{q(\theta)}{\pi_u(\theta)} d\theta = \text{KL}(q, \pi) - \log Z$ .

► In particular,  $L(q) = -J(q) \leq \log Z$ . For

$$\pi_u(\theta) = p(\text{data}|\theta)p(\theta),$$

$L(q)$  is thus a lower bound for the evidence  $p(\text{data})$  (ELBO).

- ▶ The most common approach is the mean-field approximation

$$\mathcal{Q} = \{\theta \mapsto \prod_{d=1}^D q_d(\theta_d)\}.$$

- ▶ Include all variables over which you integrate, e.g.

$$q(\theta, z_{1:n}) = \prod_{d=1}^D q_d(\theta_d) \prod_{i=1}^N q_i(z_i).$$

- ▶ Try to keep some dependence if it is key in your application.
- ▶ If your original model has NEF conditionals, **coordinate-wise maximization of  $q \mapsto L(q)$  is easy.**





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$$\begin{aligned}
 \log p(y, z, \pi, B) &= \sum_{i=1}^N \left[ \log p(\pi_i | \alpha) + \sum_{\ell=1}^{L_i} \left( \log p(z_{i\ell} | \pi_i) + \log p(y_{i\ell} | z_{i\ell}, B) \right) \right] + p(B | \gamma) \\
 &\propto \sum_{i=1}^N \left[ \sum_{k=1}^K \alpha_k \log \pi_{ik} + \sum_{\ell=1}^{L_i} \left( \sum_{k=1}^K 1_{z_{i\ell}=k} \log \pi_{ik} + \sum_{v=1}^V \sum_{k=1}^K 1_{y_{i\ell}=v} 1_{z_{i\ell}=k} \log b_{kv} \right) \right] \\
 &\quad + \sum_{k=1}^K \sum_{v=1}^V \gamma_k \log b_{kv}.
 \end{aligned}$$

## Lemma (exercise)

Let  $\Psi(\cdot) := \Gamma'(\cdot)/\Gamma(\cdot)$  be the digamma function. Then

$$\mathbb{E}_{\text{Dir}(\theta|\eta)} \log \theta_i = \Psi(\eta_i) - \Psi(\|\eta\|_1) =: \Psi_i(\eta).$$

$$\log p(y, z, \pi, B)$$

$$\propto \sum_{i=1}^N \left[ \sum_{k=1}^K \alpha_k \log \pi_{ik} + \sum_{\ell=1}^{L_i} \left( \sum_{k=1}^K 1_{z_i \ell = k} \log \pi_{ik} + \sum_{v=1}^V \sum_{k=1}^K 1_{y_i \ell = v} 1_{z_i \ell = k} \log b_{kv} \right) \right] + \sum_{k=1}^K \sum_{v=1}^V \gamma_k \log b_{kv}.$$

$$\log p(y, z, \pi, B)$$

$$\propto \sum_{i=1}^N \left[ \sum_{k=1}^K \alpha_k \log \pi_{ik} + \sum_{\ell=1}^{L_i} \left( \sum_{k=1}^K 1_{z_{i\ell}=k} \log \pi_{ik} + \sum_{v=1}^V \sum_{k=1}^K 1_{y_{i\ell}=v} 1_{z_{i\ell}=k} \log b_{kv} \right) \right] + \sum_{k=1}^K \sum_{v=1}^V \gamma_k \log b_{kv}.$$

$$\begin{aligned} \log p(y, z, \pi, B) \\ \propto \sum_{i=1}^N \left[ \sum_{k=1}^K \alpha_k \log \pi_{ik} + \sum_{\ell=1}^{L_i} \left( \sum_{k=1}^K 1_{z_i \ell = k} \log \pi_{ik} + \sum_{v=1}^V \sum_{k=1}^K 1_{y_i \ell = v} 1_{z_i \ell = k} \log b_{kv} \right) \right] + \sum_{k=1}^K \sum_{v=1}^V \gamma_v \log b_{kv}. \end{aligned}$$

- ▶ Storing  $\tilde{z}_{i\ell k}$  requires  $\mathcal{O}(NK \sum_i L_i)$  space. In practice, one works with (sparse) count data

$n_{iv}$  = number of times word  $v$  appears in document  $i$ ,

and variables  $c_{ivk}$ , thus reducing storage costs (and the dimension of the underlying integral!) to  $\mathcal{O}(NVK)$ .









- ▶ For hidden variable models, **EM** is VB with

$$q(z, \theta) = \pi(z|\theta)\delta_{\tilde{\theta}}(\theta).$$

- ▶ **Variational EM** is VB with

$$q(z, \theta) = q(z)\delta_{\tilde{\theta}}(\theta).$$

- ▶ VB for any PGMs with NEF arrows is **variational message passing**.
- ▶ Rather approximating

$$\pi(\theta) \approx \prod_{f=1}^F q_f(\theta)$$

leads to **expectation propagation**.

- ▶ These days, **ADVI with stochastic gradients** is the default VI choice in probabilistic programming software like PyMC3, Stan, or PyRo.

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Given a loss  $\ell$ , a divergence  $D$ , and a set of distributions  $\mathcal{Q}$ , consider

$$q^* \in \arg \min_{q \in \mathcal{Q}} \mathbb{E}_q \sum_{i=1}^N \ell(\theta, x_i) + D(q(\theta) d\theta, p(\theta) d\theta). \quad (P(\ell, D, \mathcal{Q}))$$

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<sup>1</sup>Knoblauch, Jewson, and Damoulas, 2022.

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<sup>2</sup>Lacoste-Julien, Huszár, and Ghahramani, 2011.

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<sup>3</sup>Teymur et al., 2021.





- ▶ Hot research topic.
- ▶ Lots of open questions on connecting VB and to the original SEU problem: is VB justified in itself or is it simply a computationally convenient backup option?
- ▶ Many partial answers, e.g. try to use VB in importance sampling, optimization-centric viewpoint.
- ▶ Meaningful candidates to replacing KL minimization: maximum mean discrepancy, PAC-Bayes bounds.

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